FUJITSU-MONAKA®

PMU Event Errata

2.0, 3 April 2026

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Contents

1 Summary of PMU Event Errata 6

2 Details of PMU Event Errata 7

2.1 Cache refill PMU Event Errata 7

2.2 MAC PMU Event Errata 8

2.3 Operation counts PMU Event Errata 9

2.4 Stall cycle PMU Event Errata 10

Revision History

Change Date	Edition	Description of Change
6/13/2025	1.0	First Release
4/3/2026	2.0	Added some events.

Introduction

This document describes the Errata information for the FUJITSU-MONAKA processor (called MONAKA, below) PMU Events.

1 Summary of PMU Event Errata

Summary of PMU Event Errata is as follows.

Table 1-1 Summary of MONAKA PMU Event Errata

PMU Event Errata Groups	Details
Cache refill	PMU Event Errata related to the L3 cache refill/hit counts
MAC	PMU Event Errata related to the MAC read request counts
Operation counts	PMU Event Errata related to the operation counts
Stall cycle	PMU Event Errata related to the stall cycle counts

The version of MONAKA can be confirmed via the MIDR_EL1 register. The correspondence between the version and the MIDR_EL1 register is shown in Table 2. Hereafter, a checkmark (✓) next to a version indicates that an Errata exist for that version of MONAKA.

Table 1-2 The correspondence between the version and the MIDR_EL1 register

MONAKA version	MIDR_EL1[23:20]	MIDR_EL1[3:0]
A	0x0	0x0
B	0x1	0x0

2 Details of PMU Event Errata

2.1 Cache refill PMU Event Errata

MONAKA has Errata for the PMU Events regarding the L3 cache refill/hit counts when the L3 prefetch is counted. Table 2-1 shows cache refill PMU Event Errata.

Table 2-1 Cache refill PMU Event Errata

PMU domain	Event Number	Event Name	Details	Version A	Version B
Core	0x0037	LL_CACHE_MISS_RD	This event may count more than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x400B	L3D_CACHE_LMISS_RD	This event may count more than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x0396	L2D_CACHE_REFILL_L3D_MISS	This event may count more than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x039A	L2D_CACHE_REFILL_L3D_MISS_PRF	This event may count more than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x039B	L2D_CACHE_REFILL_L3D_MISS_HWPRF	This event may count more than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x039C	L2D_CACHE_REFILL_L3D_HIT	This event may count less than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x03A0	L2D_CACHE_REFILL_L3D_HIT_PRF	This event may count less than the expected value when the L3 cache prefetch is counted.	✓	✓
Core	0x03A1	L2D_CACHE_REFILL_L3D_HIT_HWPRF	This event may count less than the expected value when the L3 cache prefetch is counted.	✓	✓

2.2 MAC PMU Event Errata

MONAKA has Errata for the PMU Events regarding MAC read request counts. Table 2-2 shows MAC PMU Event Errata.

Table 2-2 MAC PMU Event Errata

PMU domain	Event Number	Event Name	Details	Version A	Version B
Uncore-MAC	0x017	MAC_READ_WAIT	This event may be counted inaccurately.	✓	

2.3 Operation counts PMU Event Errata

MONAKA has Errata for the PMU Events regarding operation counts. Table 2-3 shows operation counts PMU Event Errata.

Table 2-3 Operation counts PMU Event Errata

PMU domain	Event Number	Event Name	Details	Version A	Version B
Core	0x80C0	FP_SCALE_OPS_SPEC	This event may count SVE FDOT (4-way, indexed) and SVE FDOT (4-way, vectors) operations less than the expected value.	✓	✓
Core	0x80C4	FP_SP_SCALE_OPS_SPEC	This event may count SVE FDOT (4-way, indexed) and SVE FDOT (4-way, vectors) operations less than the expected value.	✓	✓
Core	0x808C	FP_FP8_SCALE_MIN_OPS_SPEC	This event may count SVE FDOT (4-way, indexed) and SVE FDOT (4-way, vectors) operations less than the expected value.	✓	✓
Core	0x80C1	FP_FIXED_OPS_SPEC	This event may count v9 FDOT (4-way, by element) and v9 FDOT (4-way, vector) operations less than the expected value.	✓	✓
Core	0x80C5	FP_SP_FIXED_OPS_SPEC	This event may count v9 FDOT (4-way, by element) and v9 FDOT (4-way, vector) operations less than the expected value.	✓	✓
Core	0x8484	FP_FP8_FIXED_MIN_OPS_SPEC	This event may count v9 FDOT (4-way, by element) and v9 FDOT (4-way, vector) operations less than the expected value.	✓	✓
Core	0x80C8	INT_SCALE_OPS_SPEC	This event may count SVE ADCLB, SVE ADCLT, SVE SBCLB and SVE SBCLT operations less than the expected value.	✓	✓

2.4 Stall cycle PMU Event Errata

MONAKA has Errata for the PMU Events regarding stall cycle counts. Table 2-4 shows stall cycle PMU Event Errata.

Table 2-4 Stall cycle PMU Event Errata

PMU domain	Event Number	Event Name	Details	Version A	Version B
Core	0x8159	STALL_FRONTEND_L1I	This event may count more than the expected value.	✓	
Core	0x815A	STALL_FRONTEND_L2I	This event may count less than the expected value.	✓	
Core	0x815B	STALL_FRONTEND_MEM	This event may count less than the expected value.	✓	
Core	0x8166	STALL_BACKEND_L2D	This event may count more than the expected value.	✓	